THE LARGEST OPEN ON WHICH A TAU-LOCAL PROPERTY OF MORPHISMS HOLDS, AND ITS COMPATIBILITY WITH FLAT FINITELY PRESENTED BASE CHANGE ON THE SOURCE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The two concluding assertions are compressed into the single sentence that they follow immediately from the construction; the construction is fully displayed and both assertions are recorded as obligations of the card.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Locality of a property of morphisms on the source in a topology. Same chapter, Definition definition-property-morphisms-local-source and its variants, used at proof lines 6911-6913. Turns a covering of W by morphisms whose composites with f have the property into the statement that the restriction of f to W has it.
- Flat, finitely presented, syntomic, smooth and etale morphisms are open. Morphisms, Lemma morphisms-lemma-fppf-open, invoked at proof lines 6909-6911. Makes each of the images whose union defines W an open subset, so that W is open.

Source reference. Original lemma 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $\tau \in \{fppf, syntomic, smooth, étale\}$. Let $\mathcal{P}$ be a property of morphisms which is τ local on the source. For any morphism of schemes $f : X \to Y$ there exists a largest open $W(f) \subset X$ such that the restriction $f|_{W(f)} : W(f) \to Y$ has $\mathcal{P}$. Moreover, if $g : X' \to X$ is flat and locally of finite presentation, syntomic, smooth, or étale and $f' = f \circ g : X' \to Y$, then $g^{-1}(W(f)) = W(f')$.*

Proof. Consider the union W of the images $g(X') \subset X$ of morphisms $g : X' \to X$ with the properties:

(1) g is flat and locally of finite presentation, syntomic, smooth, or étale, and

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(2) the composition $X' \to X \to Y$ has property $\mathcal{P}$.

Since such a morphism g is open (see Morphisms, Lemma 01UA) we see that $W \subset X$ is an open subset of X. Since $\mathcal{P}$ is local in the τ topology the restriction $f|_W : W \to Y$ has property $\mathcal{P}$ because we are given a τ covering $\{X' \to W\}$ of W such that the pullbacks have $\mathcal{P}$. This proves the existence of $W(f)$. The compatibility stated in the last sentence follows immediately from the construction of $W(f)$. $\square$